Interactive Query beyond Linear Utility Model

Supplementary Appendix

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APPENDIX A
PROOF

Proof of Lemma 4.2. Fix feasible ranges $\mathcal{R}_t \subseteq \mathcal{R}_{t-1}$. Assume that $\mathcal{R}_{t-1}$ has positive volume. Under ideal conditional sampling, the parameter pairs $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_N$ are independent and uniform in $\mathcal{R}_{t-1}$. Define $X_i = \mathbf{1}\{\boldsymbol{\theta}_i \in \mathcal{R}_t\}$ for $i \in \{1, \ldots, N\}$. Then X_i are independent Bernoulli variables with mean $r_t = V(\mathcal{R}_t)/V(\mathcal{R}_{t-1})$, and $\widehat{r}_t = N^{-1}\sum_{i=1}^{N} X_i = N_s^{(t)}/N$.

For any $\lambda > 0$, Markov's inequality and independence give

$$\begin{aligned}\Pr(\widehat{r}_t - r_t \geq \epsilon) &\leq e^{-\lambda N\epsilon}\mathbb{E}\left[e^{\lambda\sum_{i=1}^{N}(X_i - r_t)}\right] \\ &= e^{-\lambda N\epsilon}\prod_{i=1}^{N}\mathbb{E}\left[e^{\lambda(X_i - r_t)}\right] \\ &\leq \exp\left[N\left(\frac{\lambda^2}{8} - \lambda\epsilon\right)\right],\end{aligned}$$

where the last inequality follows from Hoeffding's lemma [1], since $X_i \in [0,1]$. The exponent is minimized when $\lambda/4 - \epsilon = 0$, giving $\lambda^\star = 4\epsilon$. Hence $\Pr(\widehat{r}_t - r_t \geq \epsilon) \leq e^{-2N\epsilon^2}$. Applying the same argument to the lower tail and combining both bounds yields

$$\Pr(|\widehat{r}_t - r_t| \geq \epsilon) \leq 2e^{-2N\epsilon^2}.$$

Thus $N \geq (2\epsilon^2)^{-1}\ln(2/\delta)$ is sufficient for an absolute error of at most ϵ with probability at least $1-\delta$.

For the cumulative estimate, fix a sequence of T nested feasible ranges $\Theta = \mathcal{R}_0 \supseteq \cdots \supseteq \mathcal{R}_T$, with each sampling domain $\mathcal{R}_{j-1}$ having positive volume. Define $r_j = V(\mathcal{R}_j)/V(\mathcal{R}_{j-1})$ and $\widehat{r}_j = N_s^{(j)}/N$. The true overall volume fraction is $\rho_T = V(\mathcal{R}_T)/V(\Theta) = \prod_{j=1}^{T} r_j$, and its estimate is $\widehat{\rho}_T = \prod_{j=1}^{T}\widehat{r}_j$. Under the same ideal sampling assumption at every stage, a union bound gives

$$\Pr\left(\bigcap_{j=1}^{T}\{|\widehat{r}_j - r_j| \leq \epsilon\}\right) \geq 1 - T\delta.$$

On this event, since $r_j, \widehat{r}_j \in [0,1]$,

$$\begin{aligned}|\widehat{\rho}_T - \rho_T| &= \left|\sum_{j=1}^{T}(\widehat{r}_j - r_j)\prod_{k<j}\widehat{r}_k\prod_{k>j} r_k\right| \\ &\leq \sum_{j=1}^{T}|\widehat{r}_j - r_j| \leq T\epsilon.\end{aligned}$$

Therefore, for this fixed sequence of T stages, the overall absolute error is at most $T\epsilon$ with probability at least $1 - T\delta$. The union bound does not require independence across stages; the ideal sampling condition is required within each stage. □

Proof of Lemma 4.3. Fix the feedback set $\mathcal{H}$ and condition on $N_s^{(t)} = m \geq 1$. Under ideal conditional sampling, the surviving pairs $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_m$ are independent and uniform in the current feasible range $\mathcal{R}$. Let $\widehat{\boldsymbol{p}}$ be the first pair's nearest point, using a fixed tie-breaking rule if needed. Let $\mathcal{B} \subseteq \mathcal{R}$ contain the parameter pairs whose nearest point differs from $\widehat{\boldsymbol{p}}$, and write $\alpha = V(\mathcal{B})/V(\mathcal{R})$. Let A be the event that all m pairs identify $\widehat{\boldsymbol{p}}$ as their nearest point.

Conditioned on $\boldsymbol{\theta}_1$, the point $\widehat{\boldsymbol{p}}$ and its disagreement region $\mathcal{B}$ are fixed. The remaining $m-1$ pairs all avoid $\mathcal{B}$ with probability $(1-\alpha)^{m-1}$. Averaging over the first surviving pair therefore gives

$$\begin{aligned}&\Pr(A \cap \{\alpha > \epsilon\} \mid \mathcal{H}, N_s^{(t)} = m) \\ &\quad = \mathbb{E}\left[\mathbf{1}\{\alpha > \epsilon\}(1-\alpha)^{m-1} \mid \mathcal{H}, N_s^{(t)} = m\right] \\ &\quad \leq (1-\epsilon)^{m-1} \leq e^{-\epsilon(m-1)}.\end{aligned}$$

If $m \geq 1 + \epsilon^{-1}\ln(1/\delta)$, this probability is at most δ. Consequently, with probability at least $1-\delta$, agreement among the surviving pairs implies $V(\mathcal{B})/V(\mathcal{R}) \leq \epsilon$. □

Proof of Lemma 4.4. Let $n \geq 2$, $M = \binom{n}{2}$, and let $\boldsymbol{p}^\star$ be the user's unique favorite point. For each $\boldsymbol{q} \in \mathcal{D} \setminus \{\boldsymbol{p}^\star\}$, let $A_{\boldsymbol{q}}$ be the event that the pair $\{\boldsymbol{p}^\star, \boldsymbol{q}\}$ has not been compared in the first t questions. Since each question independently selects a distinct-point pair uniformly from the M possible pairs, $\Pr(A_{\boldsymbol{q}}) = (1 - 1/M)^t$. A union bound over the $n-1$ other points gives

$$\begin{aligned}\Pr\left(\bigcup_{\boldsymbol{q}\neq\boldsymbol{p}^\star} A_{\boldsymbol{q}}\right) &\leq (n-1)(1-1/M)^t \\ &\leq (n-1)e^{-t/M}.\end{aligned}$$

For $t = \lceil M\ln((n-1)/\delta)\rceil$, this probability is at most δ.

Under noiseless feedback, comparing $\boldsymbol{p}^\star$ with $\boldsymbol{q}$ adds the constraint $f(\boldsymbol{p}^\star) < f(\boldsymbol{q})$. Once all $n-1$ such comparisons have been made, every parameter pair consistent with the collected feedback satisfies these inequalities for every $\boldsymbol{q} \neq \boldsymbol{p}^\star$. Thus, with probability at least $1-\delta$, all feasible parameter pairs identify $\boldsymbol{p}^\star$ as their nearest point.

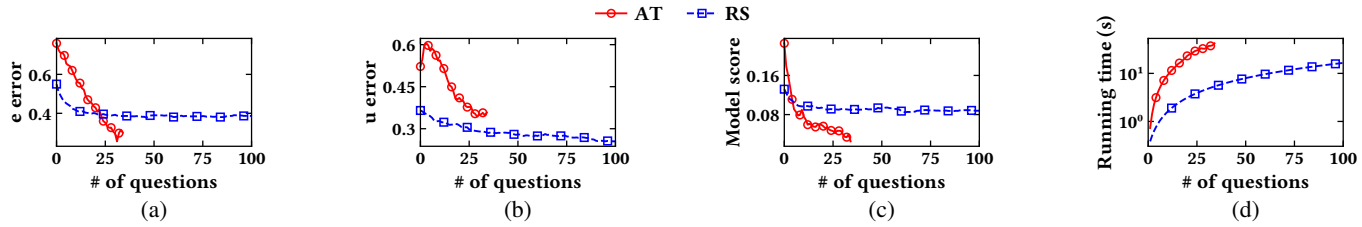


Fig. C.1. Learning behavior on Car: (a) $\boldsymbol{e}$ error; (b) $\boldsymbol{u}$ error; (c) model score; (d) cumulative running time.

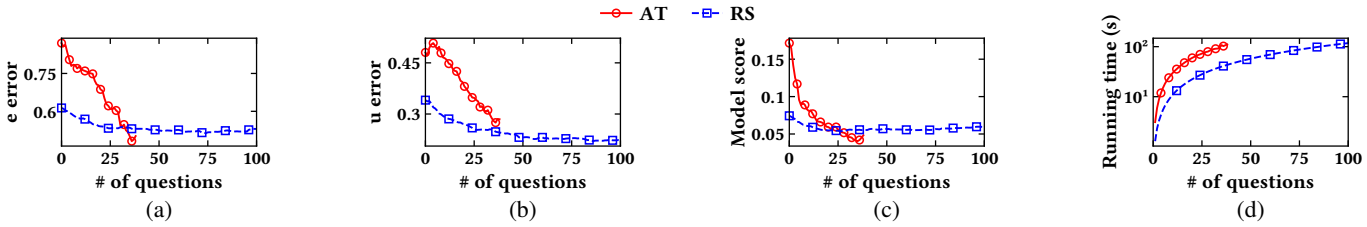


Fig. C.2. Learning behavior on Flight: (a) $\boldsymbol{e}$ error; (b) $\boldsymbol{u}$ error; (c) model score; (d) cumulative running time.

This conclusion concerns the entire feasible range and is independent of the sampling procedure. □

Proof of Lemma 5.5. Since $\eta_{\boldsymbol{q}_j,\boldsymbol{p}_i} < \eta_{\boldsymbol{q}_{j-1},\boldsymbol{q}_j}$, point $\boldsymbol{p}_i$ is closer than $\boldsymbol{q}_j$ to any $(\boldsymbol{e}(\eta),\boldsymbol{u}(\eta))$ with $\eta \in I_{\boldsymbol{q}_j}$. By definition, for any $(\boldsymbol{e}(\eta),\boldsymbol{u}(\eta))$ with $\eta \in I_{\boldsymbol{q}_j}$, point $\boldsymbol{q}_j$ is the nearest point among $\mathcal{S}_{i-1}$. Thus, for any $(\boldsymbol{e}(\eta),\boldsymbol{u}(\eta))$ with $\eta \in I_{\boldsymbol{q}_j}$, point $\boldsymbol{p}_i$ is the nearest point among $\mathcal{S}_i$. □

Proof of Lemma 5.6. Let $\eta^* = \eta_{\boldsymbol{q}_j,\boldsymbol{p}_i}$. During the backward scan, $\boldsymbol{p}_i$ is the unique score minimizer for sufficiently large η. For each $\boldsymbol{q} \in S_{i-1}$, the difference $d_{\boldsymbol{q}}(\eta) = g_\eta(\boldsymbol{q}) - g_\eta(\boldsymbol{p}_i)$ is affine. The insertion order gives $d_{\boldsymbol{q}}(0) \le 0$, whereas $d_{\boldsymbol{q}}(\eta) > 0$ for sufficiently large η. Consequently, each $d_{\boldsymbol{q}}$ has a strictly positive slope. Since η^* lies inside $I_{\boldsymbol{q}_j}$, $\boldsymbol{q}_j$ minimizes the old scores there and ties with $\boldsymbol{p}_i$ at η^*. Thus $d_{\boldsymbol{q}}(\eta^*) \ge 0$ for every $\boldsymbol{q} \in S_{i-1}$. For $\eta > \eta^*$, the positive slopes give $d_{\boldsymbol{q}}(\eta) > 0$ for every old tuple, so $\boldsymbol{p}_i$ is the unique score minimizer over S_i. For $0 \le \eta < \eta^*$, we have $d_{\boldsymbol{q}_j}(\eta) < 0$. The old minimum is therefore strictly smaller than $g_\eta(\boldsymbol{p}_i)$, so the selected tuple remains unchanged. At η^*, $\boldsymbol{q}_j$ and $\boldsymbol{p}_i$ both attain the minimum; the fixed priority order determines which tuple is selected. □

Proof of Theorem 5.8. Since there are n points, we first need $O(n \log n)$ time to sort all the points. Then, consider the step that we add point $\boldsymbol{p}_i$ to $\mathcal{I}$ and $\mathcal{P}$. Suppose the scan stops at point $\boldsymbol{q}_j$. We need $O(1)$ time to scan each of the points $\boldsymbol{q}_j, \boldsymbol{q}_{k-1}, ..., \boldsymbol{q}_{j+1}$. This can be charged to the replacements of $\boldsymbol{q}_k, \boldsymbol{q}_{k-1}, ..., \boldsymbol{q}_{j+1}$. We also spend $O(1)$ time at $\boldsymbol{q}_j$. This can be charged to the addition of $\boldsymbol{p}_i$. Since a point is added and replaced at most once in the process, we need $O(1)$ time for each point. Because there are n points, we need $O(n)$ time. Thus, the time complexity is $O(n \log n)$ in total. □

Appendix B
Stopping Condition of AT

We show that the stopping condition in AT can be checked exactly by solving linear programs. Suppose that the current anchor is $(\boldsymbol{e},\boldsymbol{u})$, and let $\boldsymbol{p}$ be its nearest point. To decide whether $\boldsymbol{p}$ can be safely returned, we need to check whether there exists another point $\boldsymbol{q} \in \mathcal{D}$ and a feasible parameter pair consistent with all collected feedback such that $\boldsymbol{q}$ is closer than $\boldsymbol{p}$. Let the feedback set be $\mathcal{H} = \{(\boldsymbol{p}_k \succ \boldsymbol{q}_k)\}_{k=1}^m$, where $\boldsymbol{p}_k \succ \boldsymbol{q}_k$ means that the user prefers $\boldsymbol{p}_k$ to $\boldsymbol{q}_k$. Each feedback induces the constraint $g(\boldsymbol{p}_k) < g(\boldsymbol{q}_k)$, where $g(\boldsymbol{p}) = \sum_{i=1}^d u[i](p[i] - e[i])^2$ is the squared weighted distance. For any two points $\boldsymbol{a}$ and $\boldsymbol{b}$, we have $g(\boldsymbol{a}) - g(\boldsymbol{b}) = \sum_{i=1}^d u[i](a[i]^2 - b[i]^2) - 2\sum_{i=1}^d u[i]e[i](a[i]-b[i])$. By introducing an auxiliary variable $z[i] = u[i]e[i]$, this difference becomes linear in $(\boldsymbol{u},\boldsymbol{z})$: $g(\boldsymbol{a}) - g(\boldsymbol{b}) = \sum_{i=1}^d u[i](a[i]^2 - b[i]^2) - 2\sum_{i=1}^d z[i](a[i]-b[i])$. Assuming $e[i] \in [0,1]$, $u[i] \ge 0$, and $\sum_{i=1}^d u[i] = 1$, the relation $z[i] = u[i]e[i]$ is equivalently represented by $0 \le z[i] \le u[i]$. Therefore, for each candidate point $\boldsymbol{q} \in \mathcal{D}$, we solve the following linear program:

$$
\begin{aligned}
\min_{\boldsymbol{u},\boldsymbol{z}} \quad & \sum_{i=1}^d u[i](q[i]^2 - p[i]^2) - 2\sum_{i=1}^d z[i](q[i]-p[i]) \\
\text{s.t.} \quad & \sum_{i=1}^d u[i](p_k[i]^2 - q_k[i]^2) - 2\sum_{i=1}^d z[i](p_k[i]-q_k[i]) \\
& \quad \le 0, \quad \forall k \in [1,m], \\
& 0 \le z[i] \le u[i], \quad \forall i \in [1,d], \\
& u[i] \ge 0, \quad \forall i \in [1,d], \\
& \sum_{i=1}^d u[i] = 1.
\end{aligned}
$$

The objective is exactly $g(\boldsymbol{q}) - g(\boldsymbol{p})$ under the transformed variables, and the constraints encode all collected feedback. If the optimal value is non-negative for every $\boldsymbol{q} \in \mathcal{D}$, then no feasible pair can make any point closer than $\boldsymbol{p}$. Since the true user preference must satisfy all collected feedback, $\boldsymbol{p}$ is guaranteed to be the user's favorite point and can be returned by AT. If the optimal value is negative for some $\boldsymbol{q}$, then there exists a feasible pair under which $\boldsymbol{q}$ is closer than $\boldsymbol{p}$; hence, $\boldsymbol{p}$ cannot be safely returned. In this case, the optimal solution

provides a feasible alternative direction for further exploration. After solving the linear program, a corresponding expected tuple can be recovered by setting $e[i] = z[i]/u[i]$ when $u[i] > 0$; when $u[i] = 0$, the value of $e[i]$ can be chosen arbitrarily within its domain because that dimension has zero weight.

APPENDIX C
ADDITIONAL LEARNING-BEHAVIOR RESULTS

Figures C.1 and C.2 show the learning trajectories on Car and Flight, complementing the Island and NBA results in Section VI-D of the main paper. They use the same experimental setup, parameter readouts, and averaging rules described there. On Car, AT's expected-tuple error and model score decrease after relatively few questions.

Across Car, Flight, Island, and NBA, AT asks between 13.50 and 31.21 questions on average before terminating, compared with 200.73–254.67 for RS. Meanwhile, RS has lower cumulative running time after the same number of questions. AT therefore reduces the number of questions required for identification while incurring greater computational cost per question in these experiments.

REFERENCES

[1] J. M. Phillips, "Chernoff-hoeffding inequality and applications," 2013. [Online]. Available: https://arxiv.org/abs/1209.6396